

Lab - Guideline

Topics

- A. IPv6 – Intro
- B. IPv6 – Static routing

A. Commands – routing table

Netstat and route commands

```
netstat -r  
route print
```

Netsh commands

```
netsh>  
    interface ipv6  
    interface ipv6 show ...  
    interface ipv6 add address ...  
    interface ipv6 add route ...  
    interface ipv6 add dnsserver ...
```

Examples: <http://computernetworkingnotes.com/ipv6-features-concepts-and-configurations/ipv6-address.html>

IPv6 has 128-bit source and destination addresses



IPv6 header

An IPv6 address is formed by two entities: prefix and interface id

Prefix	Interface ID
3FFE:0301:DEC1::	0A00:2BFF:FE36:701E

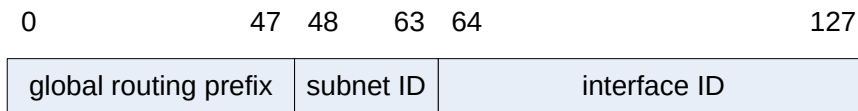
IPv6 Addressing methods:

- Unicast: one to one,
- Multicast: one to many (to a group)
- Anycast: one to nearest interface (from a group)

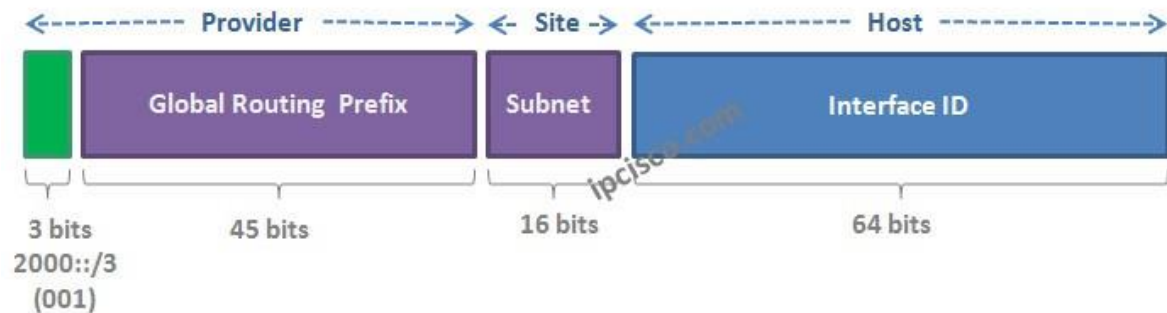
IPv6 Addresses:

Unicast Addresses

- global



Global Unicast IPv6 Address



Example: 2a03:2880:f123:83:face:b00c:0:25de

- private
 - o Link local (for a link, used for address configuration, neighbor discovery):
 - starts with FE80::/10 - FEB0::/10 prefix

Example: fe80::c891:e16d:b55:7253

- o Site local (for an organization):
 - starts with FEC0::/10 - FEF0::/10 prefix – where deprecated
- o unique local addresses (use in private IPv6 networks):
 - address block fc00::/7

Example: fde4:8dba:82e1::ea34::71ff:fe0

Adresele Multicast

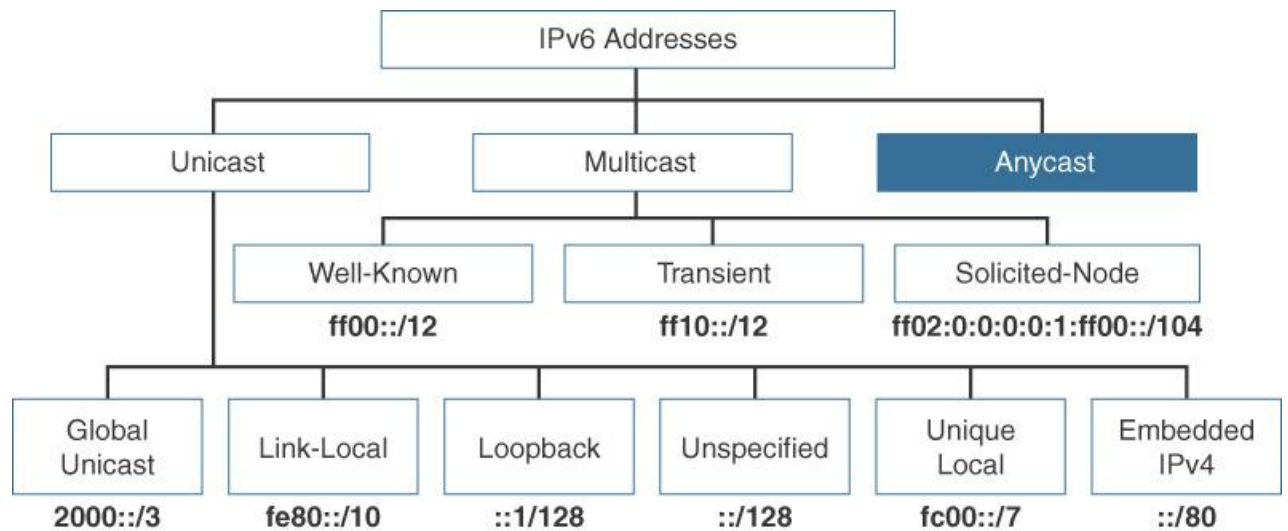
0	7	8	11	12	15	16	127
Indicator (FF)		Flags 000T (transient flag)		Scope ID		Group ID	
FF		IF T =0 well known multicast (permanent) IF T =1 transient		node-local=1, link-local=2, site-local=5, organization-local=8, global=14			

- Well-known Multicast Groups
 - o solicited-node
 - o all nodes
 - o all routers

Example: FF02::1:FF55:7253

Anycast Addresses

- “send to any one member of this group”



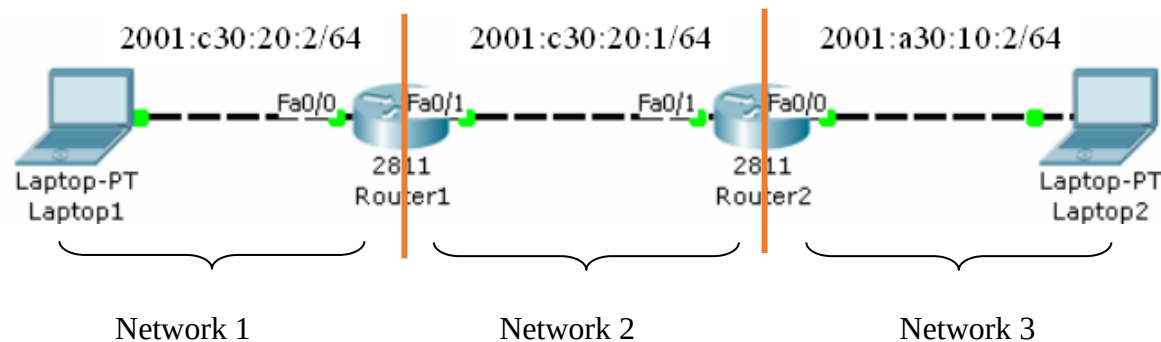
(from www.ciscopress.com)

Global addresses auto-configuration

- Stateless – **router**
 - Link-Local Address Generation
 - Link-Local Address Uniqueness Test: sends a Neighbor Solicitation message using the Neighbor Discovery (ND) protocol to check if it is unique
 - Link-Local Address Assignment
 - Router Contact
 - listening for Router Advertisement messages sent periodically by routers or
 - by sending a specific Router Solicitation to ask a router for information
 - Global Address Configuration
 -
- Statefull - **DHCPv6**
 - Using DHCPv6 protocol by deploying a DHCPv6 server in the network

B. IPv6 – static routing

Laboratory test configuration:



Step0: Enter configuration mode and configure ipv6 unicast routing (enables the forwarding of IPv6 unicast datagrams)

Router>enable

Router#configure terminal

Router(config)# ipv6 unicast-routing

Step1: Assign static IPv6 addresses to router interfaces and computers

General syntax:

Router(config-if)#ipv6 address ipv6-address/prefix-length [eui-64]

Example:

Recommended method (manually specifying the Prefix and the **eui-64 option** will assign automatically an InterfaceId):

Router1(config)#interface fastEthernet 0/0

Router1(config-if)#ipv6 address 2001:C30:20:2::/64 eui-64

Router1(config-if)#no shutdown

or (not recommended, unless you have a specific IPv6 address)

Router1(config)#interface fastEthernet 0/0

Router1(config-if)#ipv6 address 2001:C30:20:2:209:7CFF:FE4D:1501/64

Router1(config-if)#no shutdown

Step1a: Show IPv6 address of the interface fastEthernet 0/0

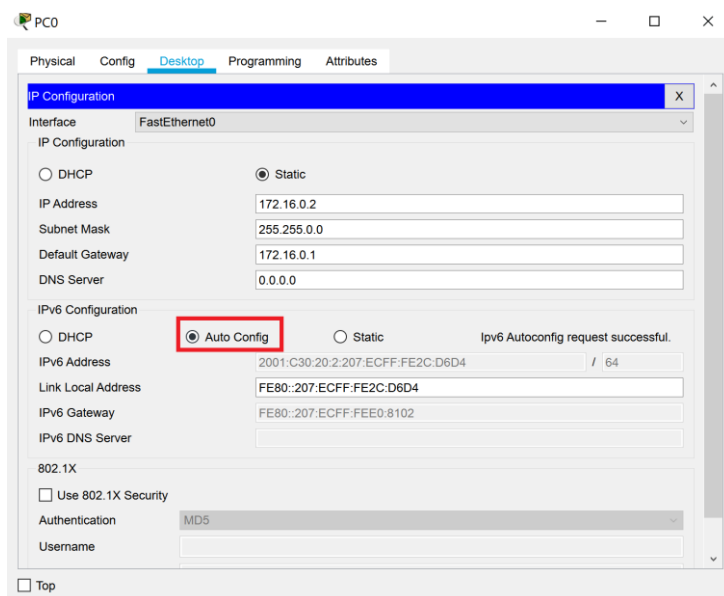
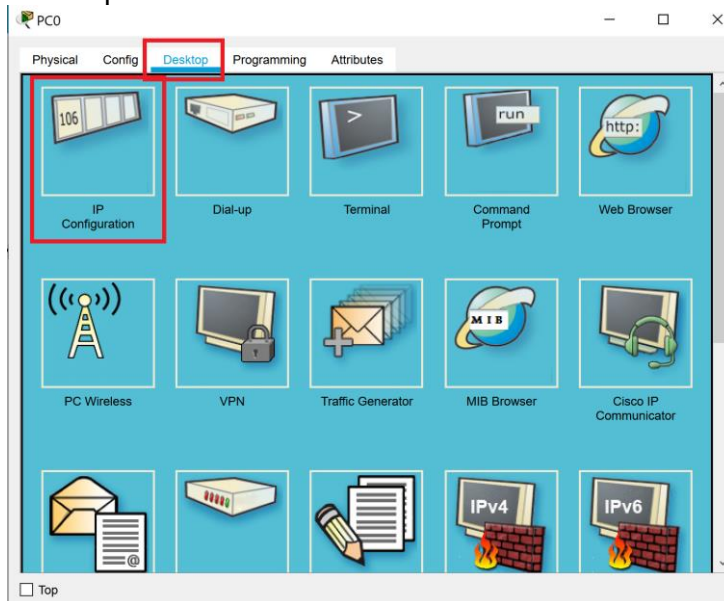
Router(config-if)#end

Router#show ipv6 interface fastEthernet 0/0

Step 1b: configure all the others interfaces on the routers with IPv6 addresses

Step2: Configure IPv6 addresses for the PCs

- On the PCs select the Auto-configuration option (stateless configuration) – see pictures below



Step3: Set static routes

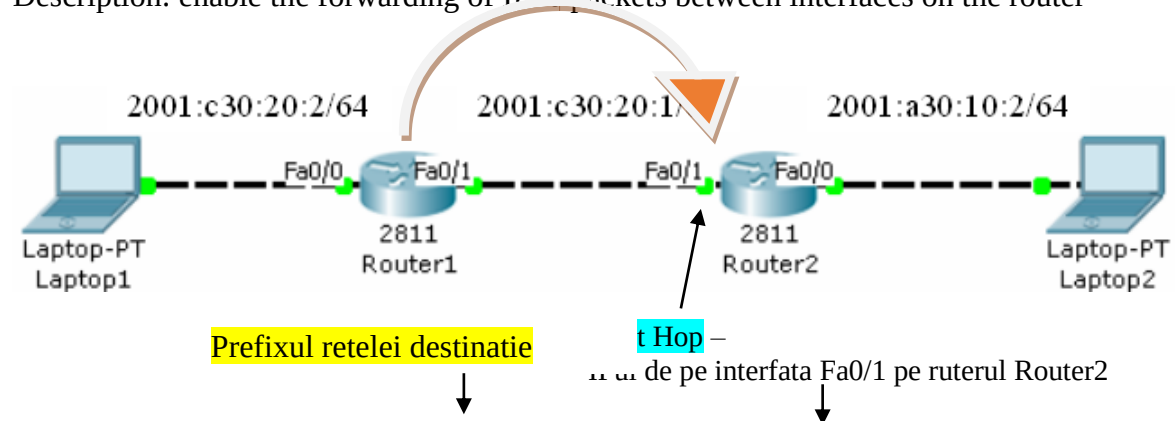
General syntax:

Router(config)#ipv6 route ipv6-prefix/prefix-length next_hop_ipv6 address/interface

Example:

Router(config)#ipv6 unicast-routing

Description: enable the forwarding of IPv6 packets between interfaces on the router



Router1(config)# ipv6 route 2001:A30:10:2::/64 2001:C30:20:1:290:2BFF:FE71:6702

Description: configure a static route to 2001:A30:10:2::/64 network

Router#show ipv6 route

Description: Visualize the routing table

Repeat step 3 for Router2 also, using the correct prefix and next-hop address

Test the connectivity.

- *ping <target IP>*
- *tracert <target IP>*